Progress report, August 31, 2026: Training and inference panels: Mehregan, Nguyen, Ravivarapu

> **Correction (October 2026):** the gate audits of October 12–14, 2026 replaced this report's Ravivarapu et al. results and its Mehregan et al. Fig. 6 results. The text below is unchanged from the original report; the current state is in the trackers: Mehregan, Nguyen, Ravivarapu.

Prepared by Devin Auch
August 31, 2026

Enhancing Adaptive Deep Brain Stimulation via Efficient Reinforcement Learning (Mehregan et al.)
Closed-Loop Neuromorphic Deep Brain Stimulation (Nguyen et al.)
Sample-Efficient Reinforcement Learning Controller for Deep Brain Stimulation in Parkinson’s Disease (Ravivarapu et al.)

Overview (August 31, 2026): Mehregan et al. replication is fully complete with all figure panels passing quantitative gates, including smoothed Fig. 4a chatter (K=3) and paired Fig. 4b training metrics. **Ravivarapu et al. (SEA-DBS)** replication is also essentially complete, with all training, frequency inference, quantization, and ablation panels (Figs 4a-7) passing quantitative gates with strict trajectory separation and aligned prefilled baselines, while leaving minor polish in the later Figs 6 and 7 for down the road. **Nguyen et al. (DSQN)** is the primary active focus, currently approximately 70% complete: GPi biomarker distribution (Fig. 3), 500-episode training dynamics (Fig. 4), and CBGT spike count / DBS energy scaling (Fig. 5) pass their quantitative gates. Current active work is centered on the later Nguyen figures: refining the training α-β power descent (which still needs about a 50-unit drop to reach full suppression), minor parameter dynamics tuning (Fig. 6), and finalizing the 50-episode evaluation rollout (Fig. 7).

Same caveats as prior reports: replications from paper text and figures (no released training / environment source), shared plant = Python reconstruction of **Kumaravelu et al. (2016)**. All replication figures include digitized ground-truth curves extracted directly from the source papers to enforce hard quantitative gates. Full gates and run commands: figures/mehregan/replications.md, figures/nguyen/replications.md, figures/ravivarapu/replications.md.

Compute upgrade: AMD Ryzen 7 5800X (8c/16t) → Ryzen 9 5900XT (16c/32t)

Report 3 listed a workstation compute upgrade as upcoming work. That upgrade is **done** (August 2026): the training workstation CPU is now an **AMD Ryzen 9 5900XT** (**16 cores** / **32 threads**), replacing the previous **AMD Ryzen 7 5800X** (**8 cores** / **16 threads**) that carried all replication trains since Report 1.

Why it matters for this project

One train at a time is not much faster. Each replication run is single-threaded (`OMP_NUM_THREADS=1`), and because integrating the biophysical network equations is pure floating-point math, one simulation run fully saturates a physical core. A single 500-episode training run still takes roughly the same wall-clock time; the upgrade buys **dedicated physical core concurrency and parallel sweep capacity**, not faster individual trains.

Area	Before: Ryzen 7 5800X (8 cores / 16 threads)	Now: Ryzen 9 5900XT (16 cores / 32 threads)
Parallel worker capacity	Reserving 2-4 cores for the OS and tooling left **4-6 concurrent simulation workers** max. A 4-variant ablation panel consumed nearly all available capacity.	Reserving 2-4 cores for the OS and tooling leaves **12-14 dedicated simulation workers** running at 100% full speed without SMT thread contention.
Multi-seed / ablation sweeps	Running a 4-variant ablation across 3 seeds (12 runs total) required queuing across multiple serial batches.	Running an entire 12-run sweep (e.g., all 4 ablation variants across 3 seeds simultaneously) in a single parallel batch.
Multitasking & eval	Running continuous parameter sweeps alongside local evaluation and plotting scripts caused sluggishness and process contention.	**Headroom to spare**: keep 12 full simulation sweeps running in the background while compiling figures, analyzing data, and running local tools.
Calendar throughput	Multi-candidate tuning was bottlenecked by serial execution lanes.	More parallel lanes mean significantly more completed hyperparameter experiments per week.

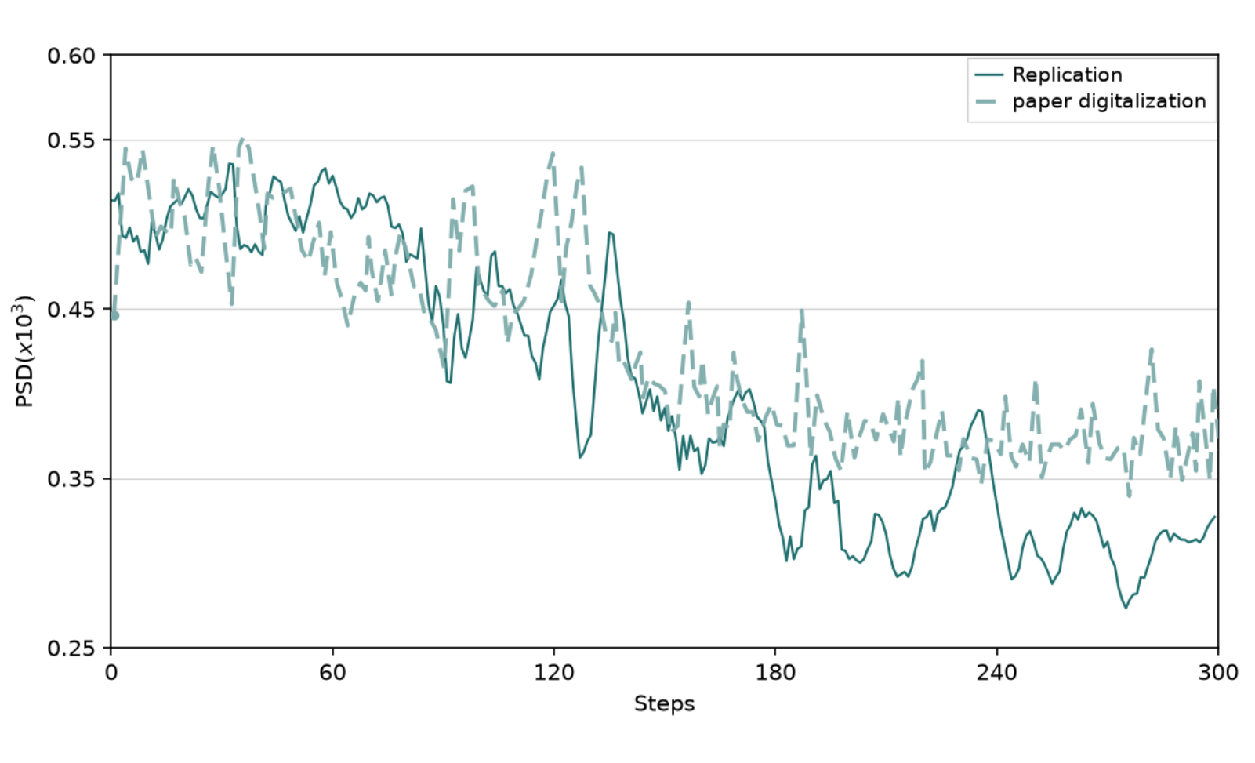
Delta since Report 3 (August 10)

Area	Report 3	Now
Compute	Planned CPU upgrade	**Done**: Ryzen 7 5800X (8c/16t) → Ryzen 9 5900XT (16c/32t)
Mehregan 4a-4b	Under revision against digitized curves	**Pass**: action persistence K=3 (Fig. 4a v38) + paired reward/PSD series (Fig. 4b v45)
Nguyen 4	In progress (shape gates)	**Pass**: 500-episode reward and episode length trajectory gates
Nguyen 5	Not in Report 3 gallery	**Pass**: CBGT spike counts and DBS energy scaling (**new in this report**)
Nguyen 6-7	Not in Report 3 gallery	**Active**: training α-β descent / parameter dynamics (Fig. 6) and 50-ep eval rollouts (Fig. 7) (**new in this report**)
Ravivarapu 4a	In progress (shape gates)	**Pass**: full digitization trajectory gates (v70)
Ravivarapu 4b-7	Not in Report 3 gallery	**Pass (Complete)**: training reward, 50/30 Hz inference, FP16 PTQ, ablation (**new in this report**)

Gallery: Mehregan et al.

Fig. 4a: Training beta vs step

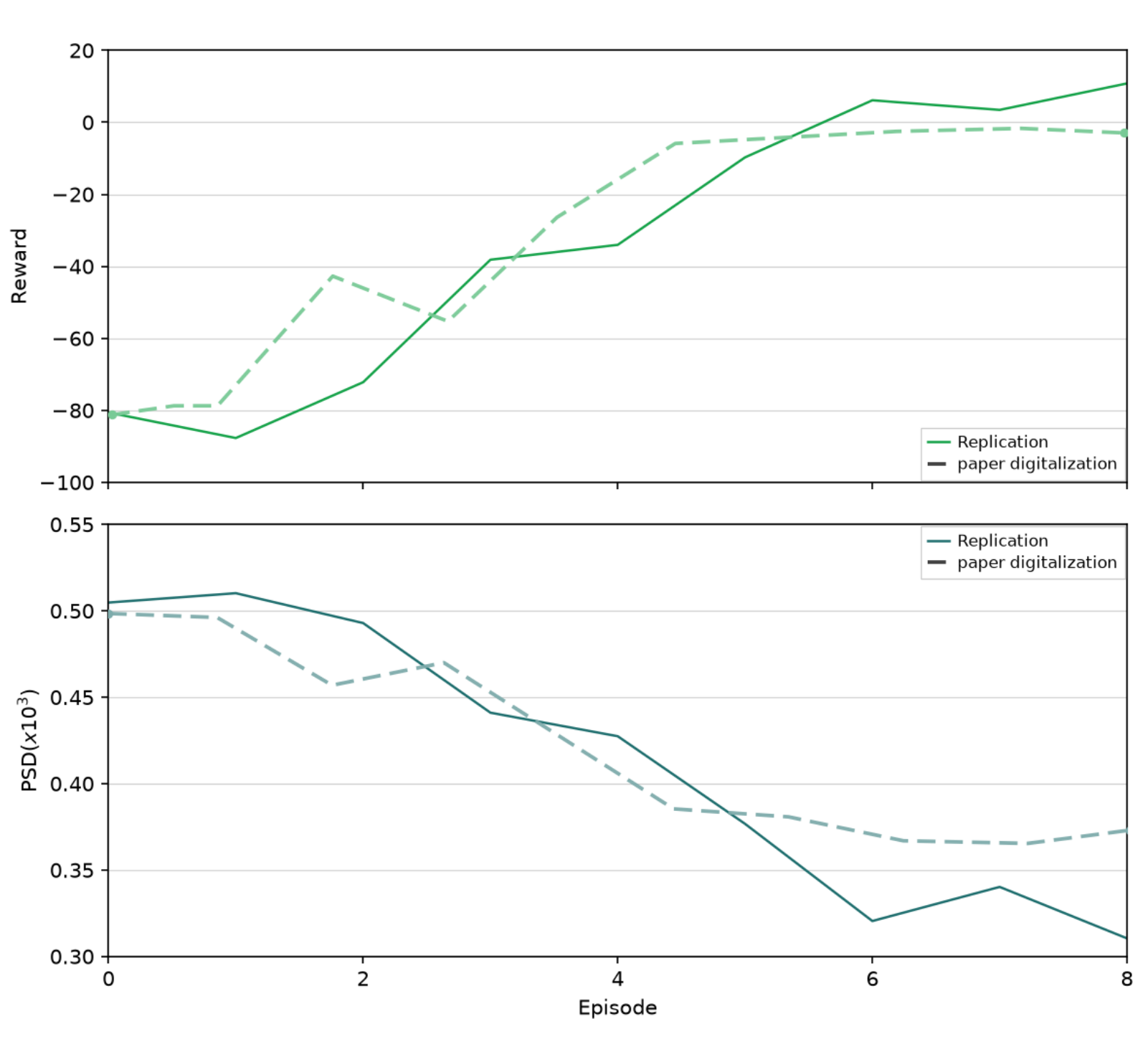
Paper claim: Noisy early β, sharp mid-run drop (~steps 130-150), lower late plateau during **45 Hz** DDPG training.



> "The average power of the beta frequency band for each step during training of the 45Hz average frequency model." *Mehregan et al.*

Fig. 4b: Training reward and episode-mean beta

Paper claim: Over episodes 0-8, reward rises toward **0** while episode-mean PSD falls (~0.50 → ~0.37); late PSD stays **above** reward threshold βt=0.35.

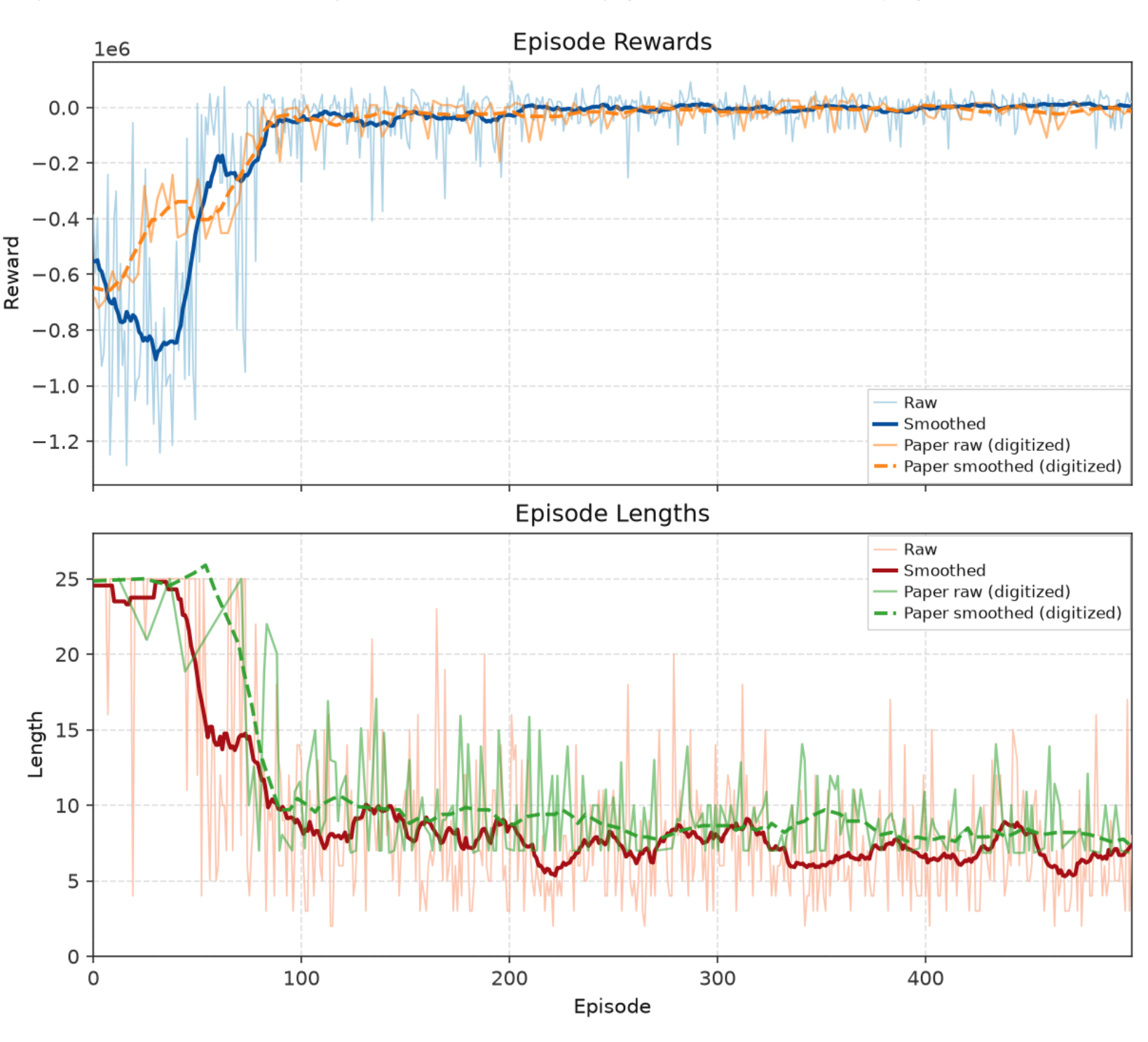


> "Beta power in GPi during each episode of training of the 45Hz (top) and accumulated reward of each training episode for the 45Hz (bottom)." *Mehregan et al.*

Gallery: Nguyen et al.

Fig. 4: Training reward + episode length

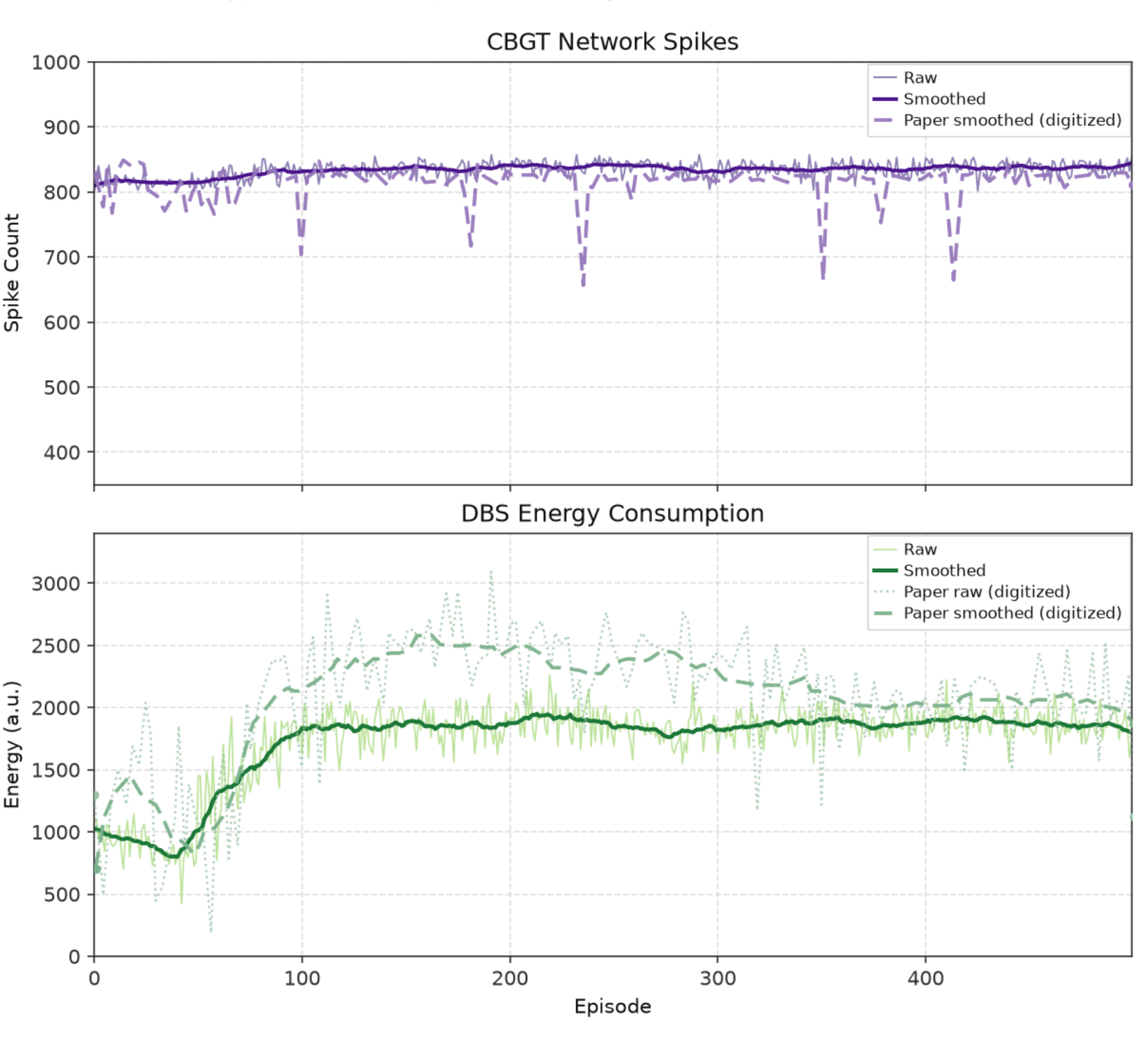
Paper claim: High early variance; smoothed reward rises toward **~0** by episodes 80-100; episode length starts at horizon (**25**), glides down mid-training, plateaus near **~8-10** by episode 100.



> "Training rewards and episode lengths over 500 episodes, showing progression from exploration to optimization." *Nguyen et al.*

Fig. 5: CBGT spikes and DBS energy

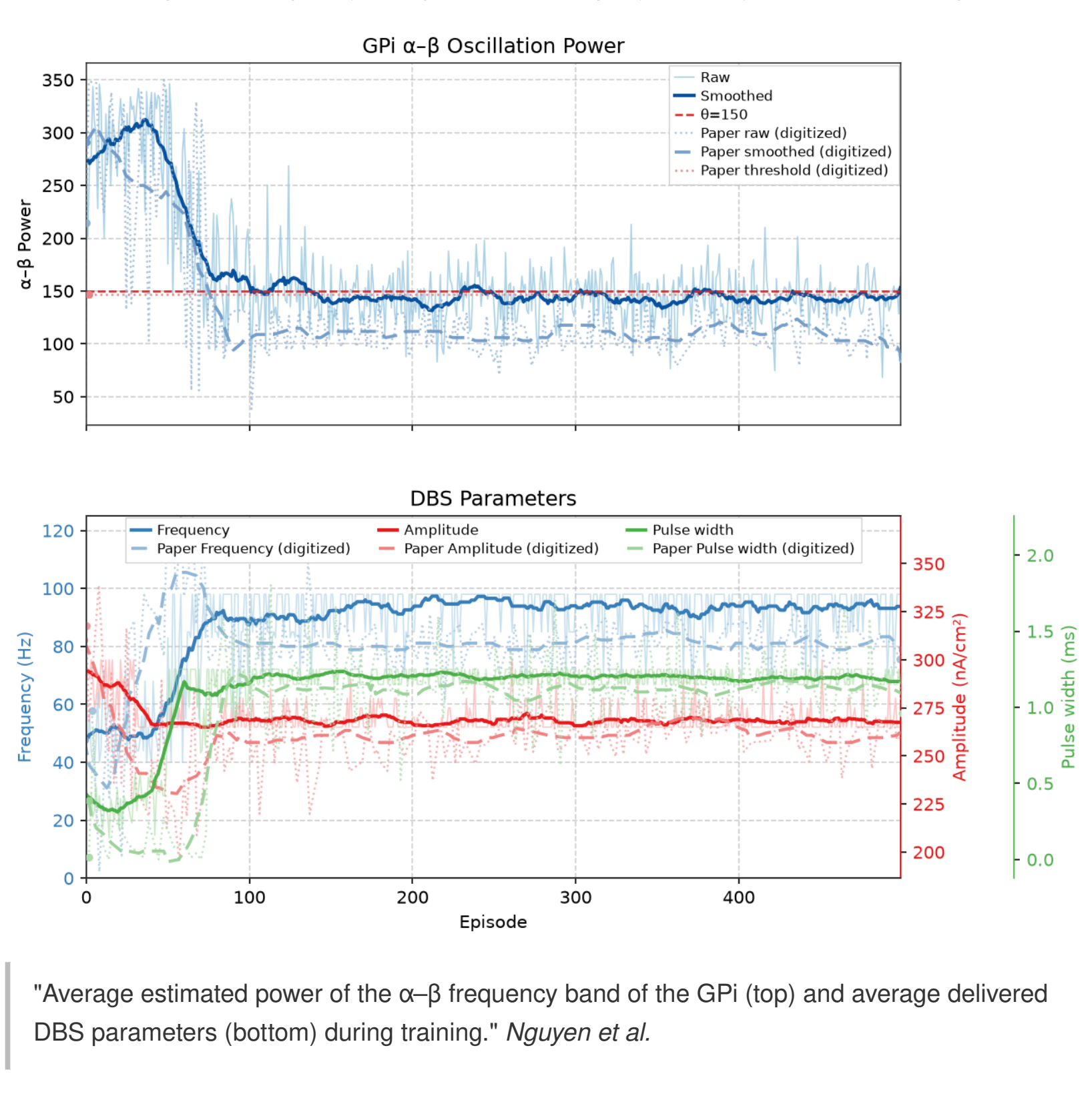
Paper claim: Basal ganglia-thalamus spike counts remain stable (~800 spikes/episode) while delivered DBS energy scales smoothly as closed-loop stimulation stabilizes.



> "Total number of spikes across the CBGT network (top) and total electrical energy delivered by DBS (bottom) per episode during training." *Nguyen et al.*

Fig. 6: α–β and DBS parameters over training

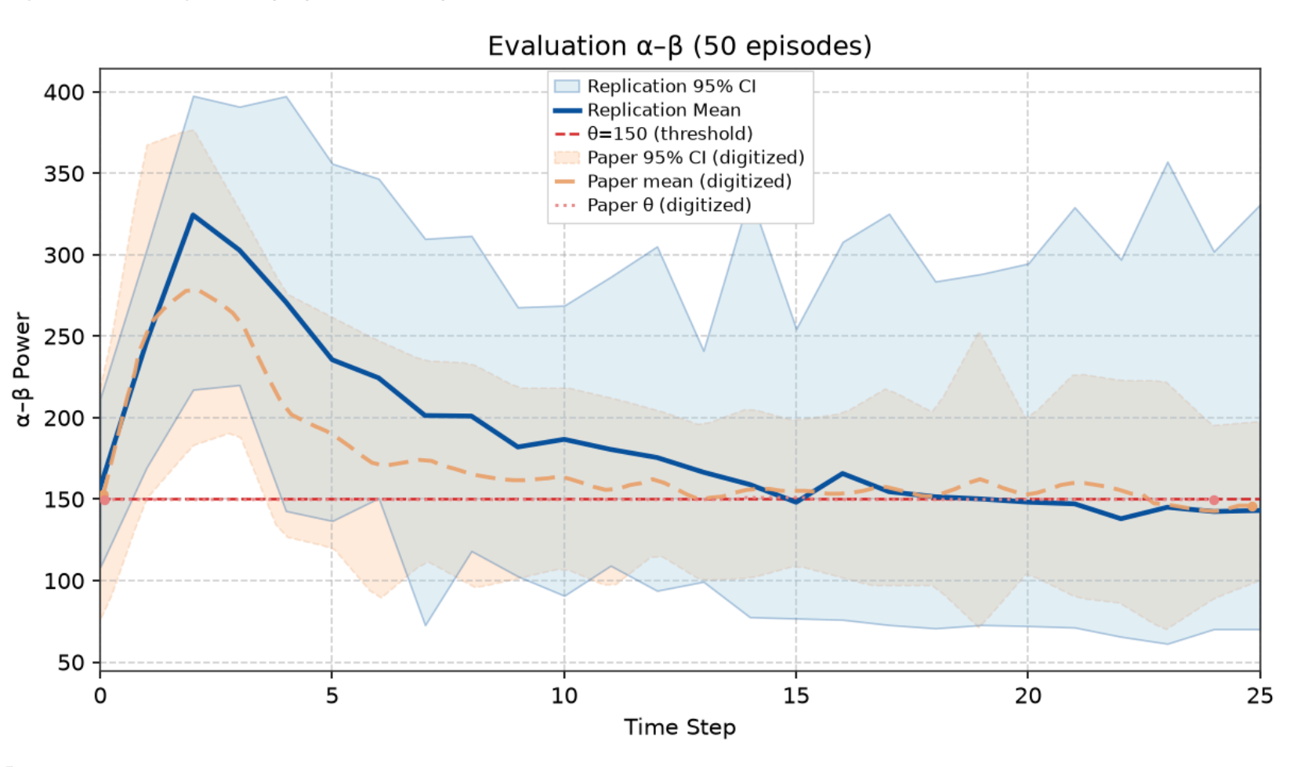
Paper claim: GPi α–β oscillation power drops below threshold θ=150 over 500 episodes while stimulation amplitude, frequency, and pulse width adapt dynamically across dedicated parameters.



> "Average estimated power of the α–β frequency band of the GPi (top) and average delivered DBS parameters (bottom) during training." *Nguyen et al.*

Fig. 7: Evaluation (50 episodes × 25 steps)

Paper claim: 50-episode evaluation rollout demonstrates rapid suppression of pathological GPi α–β oscillations within 10–15 steps, stabilizing near ~160 with confidence intervals bounded tightly against the digitized paper envelope.

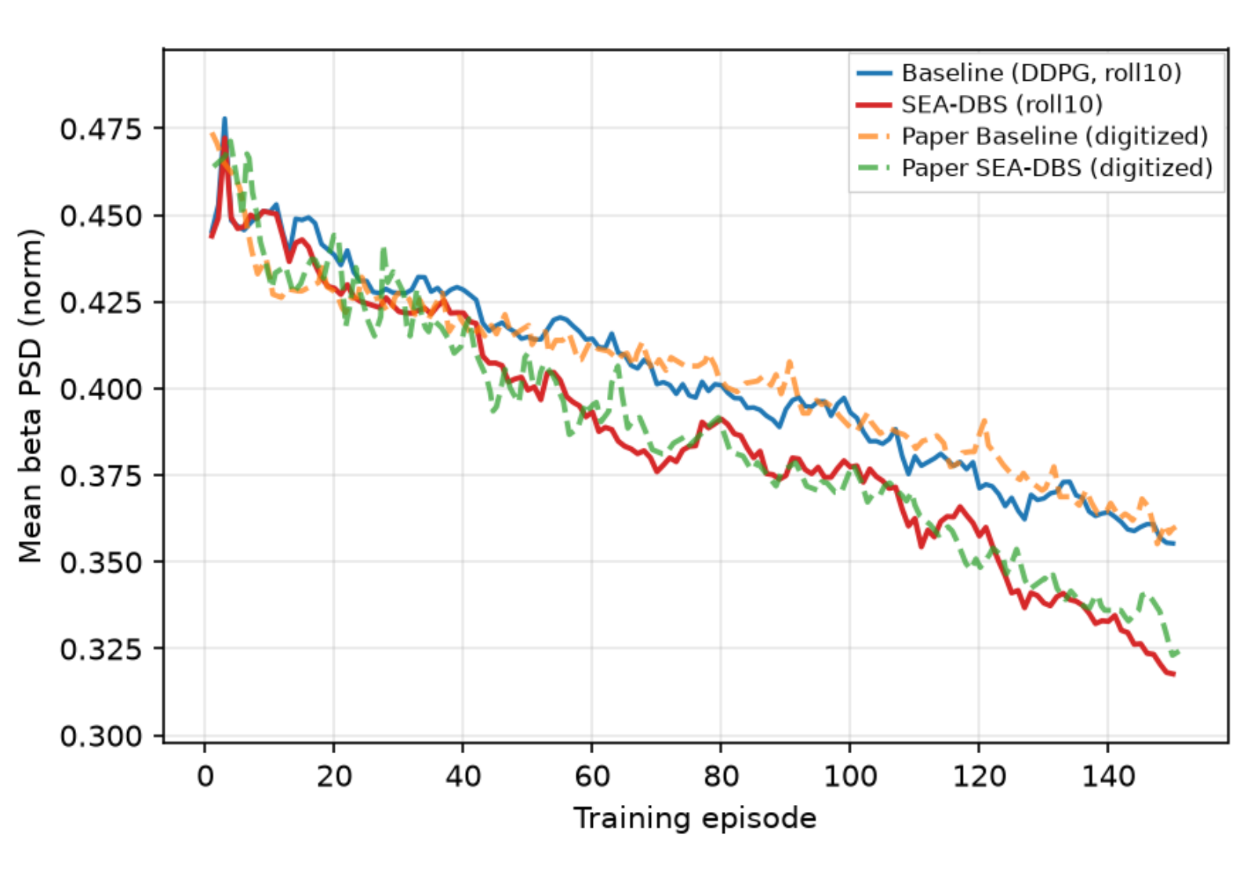


> "Evaluation of the trained DSQN controller across 50 independent episodes (25 steps each)." *Nguyen et al.*

Gallery: Ravivarapu et al.

Fig. 4a: Training PSD (Baseline vs SEA-DBS)

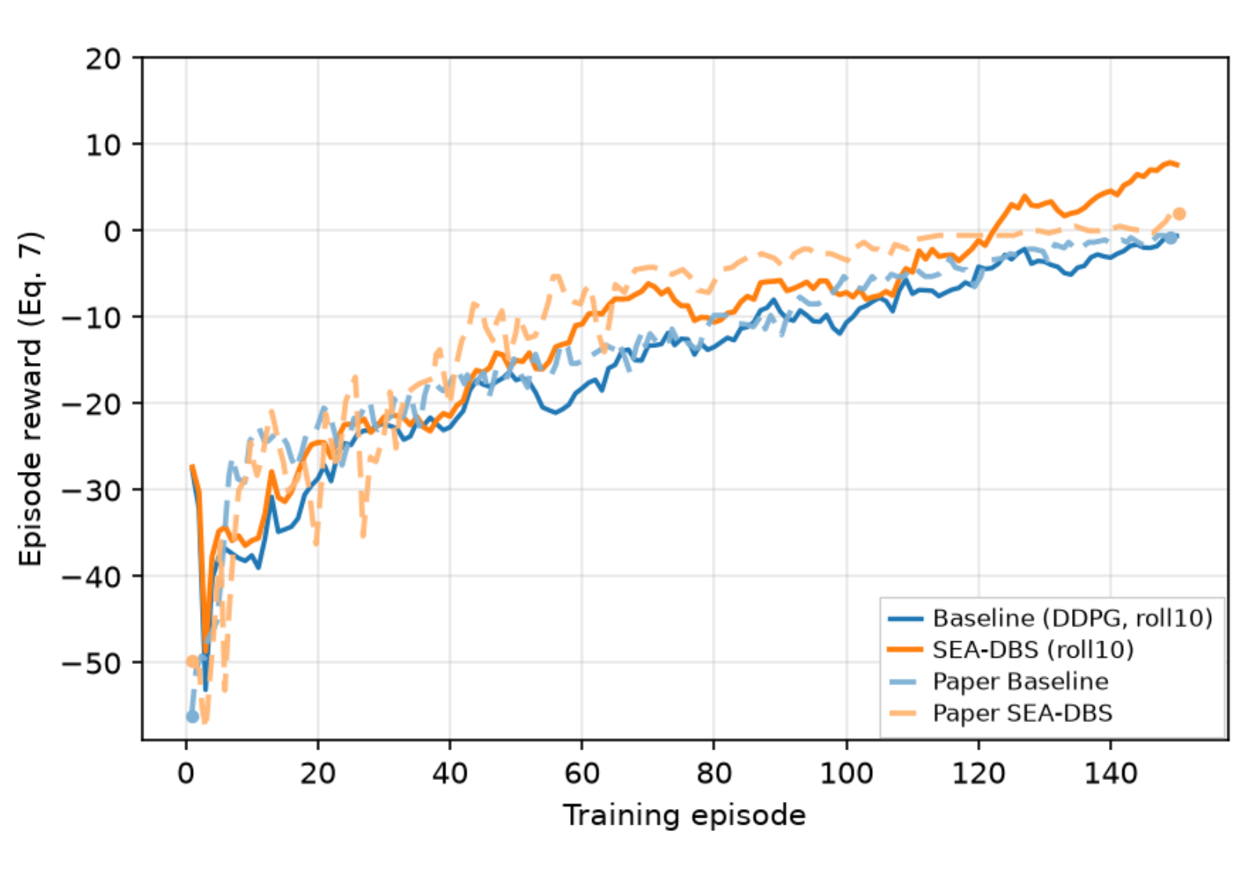
Paper claim: **SEA-DBS** shows stronger, more consistent β suppression over **150** episodes than **Baseline** DDPG.



> "PSD over training episodes (SEA-DBS vs Baseline)." *Ravivarapu et al.*

Fig. 4b: Training reward vs episode

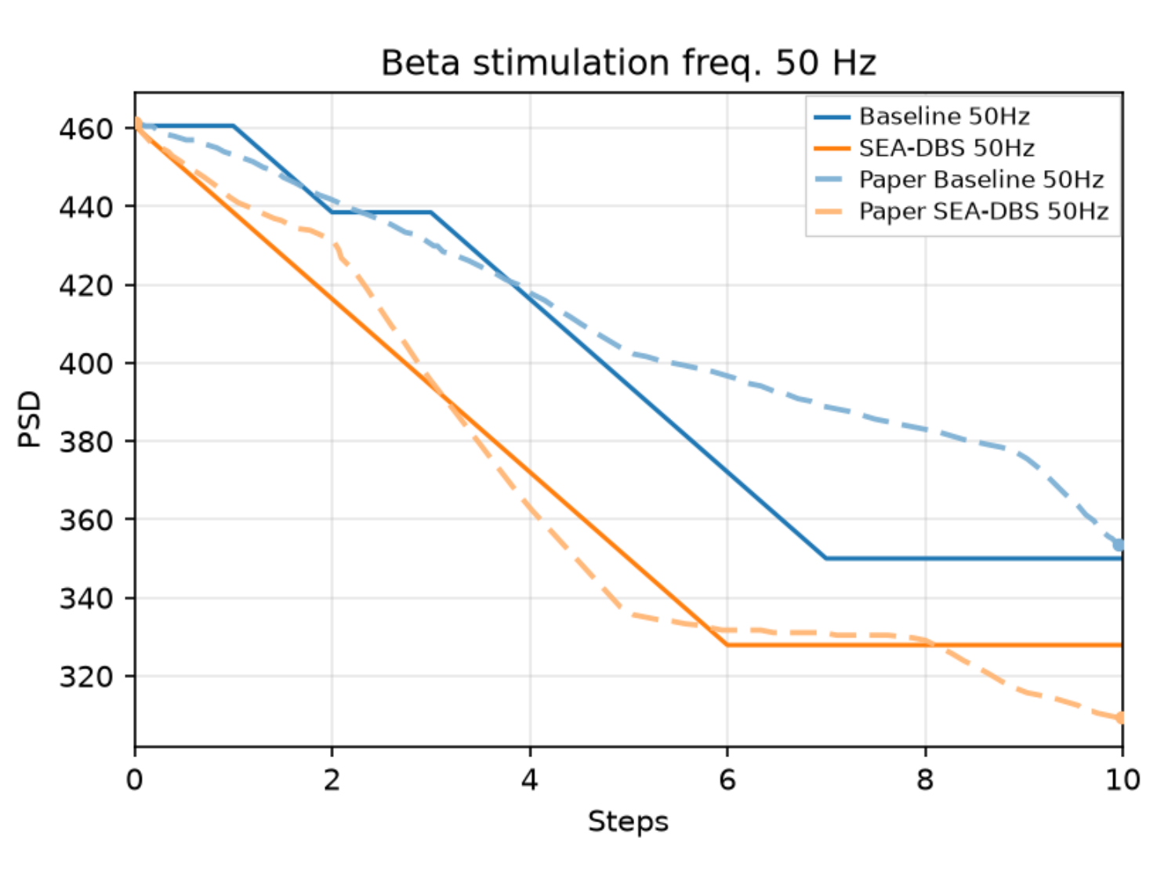
Paper claim: SEA-DBS reaches **higher** rewards with a **faster** early rise than Baseline during the same training run as Fig. 4a.



> "Cumulative reward per training episode (SEA-DBS vs Baseline)." *Ravivarapu et al.*

Fig. 5a: Inference @ 50 Hz

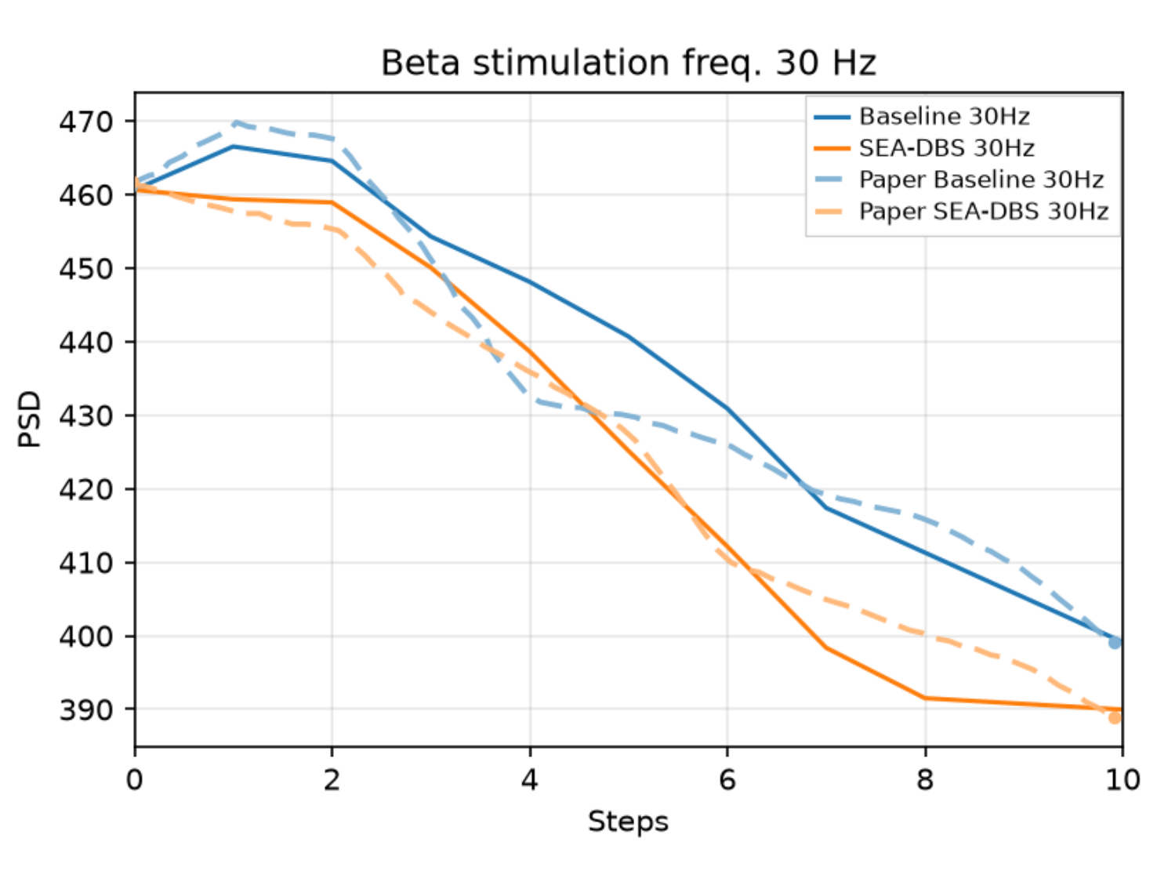
Paper claim: **50 Hz** carrier more effectively disrupts pathological oscillations than 30 Hz; SEA-DBS **below** Baseline on PSD after onset.



> "Inference GPi beta PSD vs step @ 50 Hz (SEA-DBS vs Baseline)." *Ravivarapu et al.*

Fig. 5b: Inference @ 30 Hz

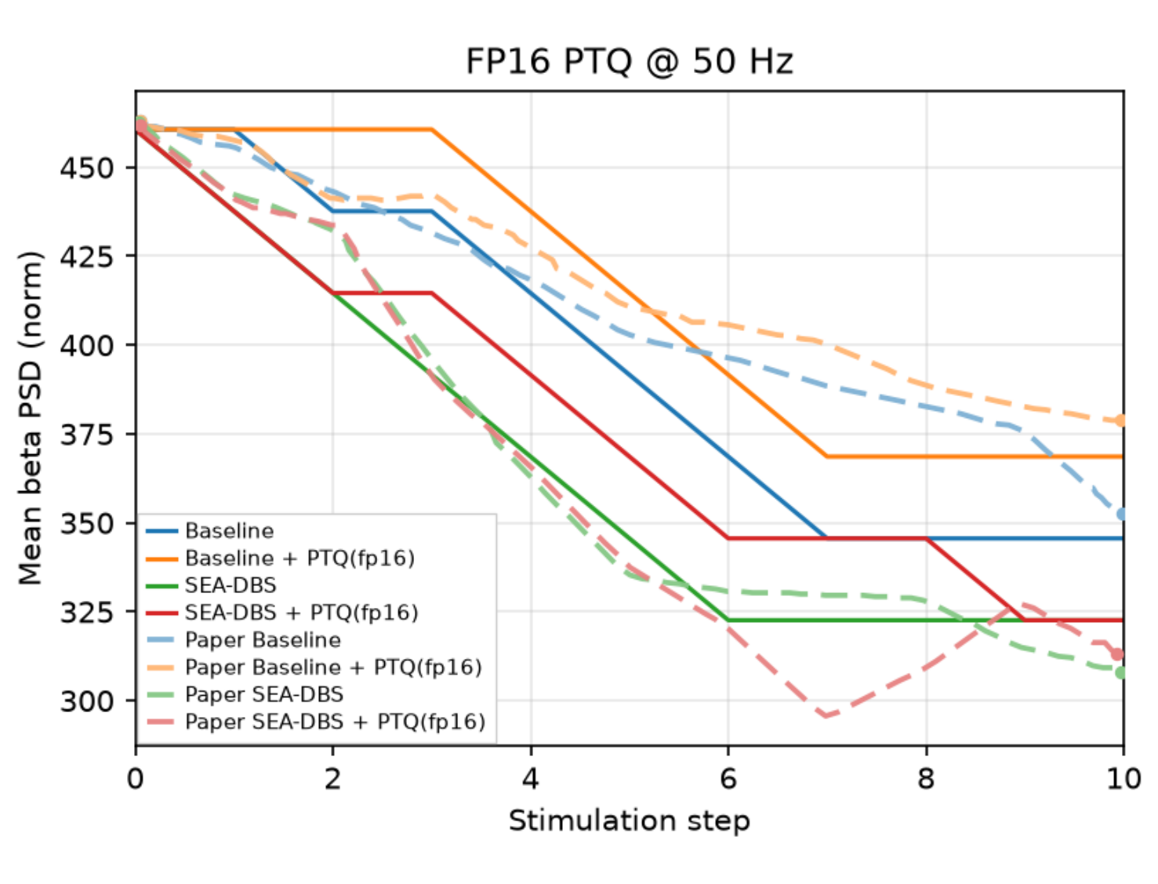
Paper claim: **30 Hz** less effective than 50 Hz; SEA-DBS still **beats** Baseline on PSD.



> "Inference GPi beta PSD vs step @ 30 Hz (SEA-DBS vs Baseline)." *Ravivarapu et al.*

Fig. 6: FP16 PTQ @ 50 Hz

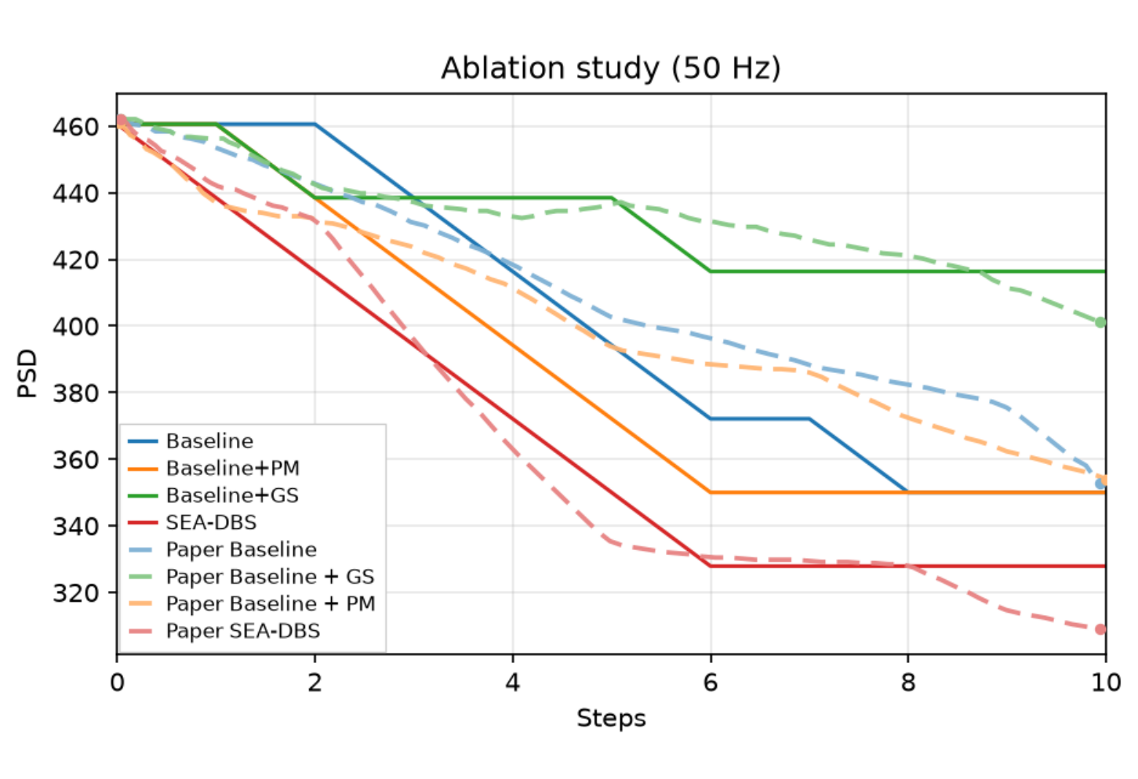
Paper claim: Post-train **fp32** vs **PTQ fp16** inference at 50 Hz; PTQ tracks fp32 suppression.



> "FP16 PTQ inference @ 50 Hz (fp32 vs PTQ)." *Ravivarapu et al.*

Fig. 7: Ablation (Baseline / +PM / +GS / SEA-DBS)

Paper claim: **+GS** and full **SEA-DBS** beat **Baseline** late; **+PM** intermediate; distinct traces per variant.



> "Ablation: Baseline, +PM, +GS, and full SEA-DBS @ 50 Hz." *Ravivarapu et al.*

What’s next

Focus	Notes
Nguyen Fig. 6 (Training α-β & parameters)	Tune SNN exploration and policy parameters to bring post-episode-100 α-β power descent down the final ~50 units below threshold θ = 150, alongside minor adjustments to stimulation parameter dynamics (amplitude, frequency, pulse width).
Nguyen Fig. 7 (50-episode eval rollout)	Roll out the finalized DSQN training checkpoint across 50 independent seeded evaluation episodes to lock in the full 25-step trajectory within the 95% confidence interval bounds.
Ravivarapu Figs. 6-7 (Deferred polish)	Revisit and apply subtle refinements to the later figures down the road after Nguyen wraps up.
Setup & environment verification	Ensure unified verification of fresh machine installation scripts (`scripts/setup.sh`), environment dependencies, and deterministic gate test suites across platforms.
Cross-controller benchmarking	Prepare unified comparative evaluations (suppression latency, energy efficiency, parameter stability) comparing Mehregan (DDPG), Nguyen (DSQN), and Ravivarapu (SEA-DBS) under identical biophysical plant conditions.
Cosmetic & aesthetic figure refinement	Polish visual presentation across all replication figures: harmonize typography, legend placement, multi-axis spacing, and color palettes to match source publication standards.
Repository architecture & organization	Clean up shared simulation code and modularize the codebase so iterating on existing controllers and replicating future papers is more streamlined.

Reproducibility & run commands

Full per-panel replication commands and gate specifications live in the repository figure trackers: - figures/mehregan/replications.md - figures/nguyen/replications.md - figures/ravivarapu/replications.md

```
git clone https://github.com/nynxlabs/rl-adaptive-dbs.git
cd rl-adaptive-dbs

# Mehregan Fig. 4a (DDPG 45 Hz training beta vs step)
uv run python -m rl_adaptive_dbs.run scripts/figures/papers/mehregan/4a/plot.py

# Ravivarapu Fig. 7 (4-variant ablation sweep at 50 Hz)
uv run python -m rl_adaptive_dbs.run scripts/figures/papers/ravivarapu/7/plot.py

# Replot any panel from cached simulation series without re-training
uv run python -m rl_adaptive_dbs.run scripts/figures/papers/ravivarapu/7/plot.py --plot-only
```

Environment note: All runs were executed and verified locally using `uv`. Standard installation and dependency management procedures are provided in `scripts/setup.sh` and the repository documentation. Note that while current workflows run cleanly against active local environments, end-to-end setup verification from a completely fresh system install hasn't been re-tested recently (tracked under upcoming work). Because full replication training sweeps involve computationally intensive biophysical ODE integration across multiple seeds and variants, reproducing all training sweeps from scratch takes a substantial amount of compute time and is recommended on an 8+ core workstation (cached series and checkpoints are provided in the repository to replot all panels instantly).

Prior reports

- **Report 1 (July 14, 2026):** Initial showcase for Mehregan et al. covering biophysical plant dynamics (Figs. 1b, 2a, 2b) and baseline 45 Hz DDPG training (Figs. 4a, 4b), establishing the continuous CBGT network simulation baseline.
- **Report 2 (July 28, 2026):** Mehregan post-train efficacy panels completed (Figs. 5a, 5b) and quantization exploration started (Figs. 6a, 6b). Initial kick-off on Nguyen et al. with Fig. 3 distribution passing and first exploratory DSQN training runs.
- **Report 3 (August 10, 2026):** Introduction of the quantitative paper-curve digitization workflow across all three papers; Mehregan Fig. 6 quantization passing on digitized gates; Nguyen Fig. 4 and Ravivarapu Fig. 4a training shape gates actively in progress.